BEING A LOCAL COMPLETE INTERSECTION MORPHISM IS SYNTOMIC LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18589-18593 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18595-18654, the complete author proof (58 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side per ruling (only A has a complete proof): the spaces twin's proof is incomplete and stays uncounted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Descent, lemma properties-morphisms-local-source; chapter lemmas flat-lci (L16202) and composition-lci (L16214); Algebra, lemma lift-syntomic; Divisors, lemma permanence-regular-immersion; Descent, lemma descending-property-regular-immersion.

Source reference. Original lemma 00T0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069F: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18111–18121. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0694: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 036H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 068Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18208–18211. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18245–18248. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18654–18654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a local complete intersection morphism" is syntomic local on the source.*

Proof. We will use the criterion of Descent, Lemma 036H to prove this. It follows from Lemmas 069K and 069J that being a local complete intersection morphism is preserved under precomposing with syntomic morphisms. It is clear from Definition 069F that being a local complete intersection morphism is Zariski local on the source and target. Hence, according to the aforementioned Descent, Lemma 036H it suffices to prove the following: Suppose $X' \to X \to Y$ are morphisms of affine schemes with $X' \to X$ syntomic and $X' \to Y$ a local complete intersection morphism. Then $X \to Y$ is a local complete intersection morphism. To see this, note that in any case $X \to Y$ is of finite presentation by Descent, Lemma 02KK. Choose a closed immersion $X \to \mathbf{A}^n_Y$. By Algebra, Lemma 00T0 we can find an affine open covering $X' = \bigcup_{i=1,\ldots,n} X'_i$ and syntomic morphisms $W_i \to \mathbf{A}^n_Y$ lifting the morphisms $X'_i \to X$, i.e., such that there are fibre product diagrams

$$\begin{array}{ccc} X'_i & \longrightarrow & W_i \\ \downarrow & & \downarrow \\ X & \longrightarrow & \mathbf{A}^n_Y \end{array}$$

After replacing X' by $\coprod X'_i$ and setting $W = \coprod W_i$ we obtain a fibre product diagram of affine schemes

$$\begin{array}{ccc} X' & \longrightarrow & W \\ \downarrow & & \downarrow h \\ X & \longrightarrow & \mathbf{A}^n_Y \end{array}$$

with $h : W \to \mathbf{A}^n_Y$ syntomic and $X' \to Y$ still a local complete intersection morphism. Since $W \to \mathbf{A}^n_Y$ is open (see Morphisms, Lemma 01UA) and $X' \to X$ is surjective we see that X is contained in the image of $W \to \mathbf{A}^n_Y$. Choose a closed

immersion $W \to \mathbf{A}^{n+m}_Y$ over $\mathbf{A}^n_Y$. Now the diagram looks like

$$\begin{array}{ccccc} X' & \longrightarrow & W & \longrightarrow & \mathbf{A}^{n+m}_Y \\ \downarrow & & \downarrow{\scriptstyle h} & \swarrow & \\ X & \longrightarrow & \mathbf{A}^n_Y & & \end{array}$$

Because h is syntomic and hence a local complete intersection morphism (see above) the morphism $W \to \mathbf{A}^{n+m}_Y$ is a Koszul-regular immersion. Because $X' \to Y$ is a local complete intersection morphism the morphism $X' \to \mathbf{A}^{n+m}_Y$ is a Koszul-regular immersion. We conclude from Divisors, Lemma 068Z that $X' \to W$ is a Koszul-regular immersion. Hence, since being a Koszul-regular immersion is fpqc local on the target (see Descent, Lemma 0694) we conclude that $X \to \mathbf{A}^n_Y$ is a Koszul-regular immersion which is what we had to show. $\square$